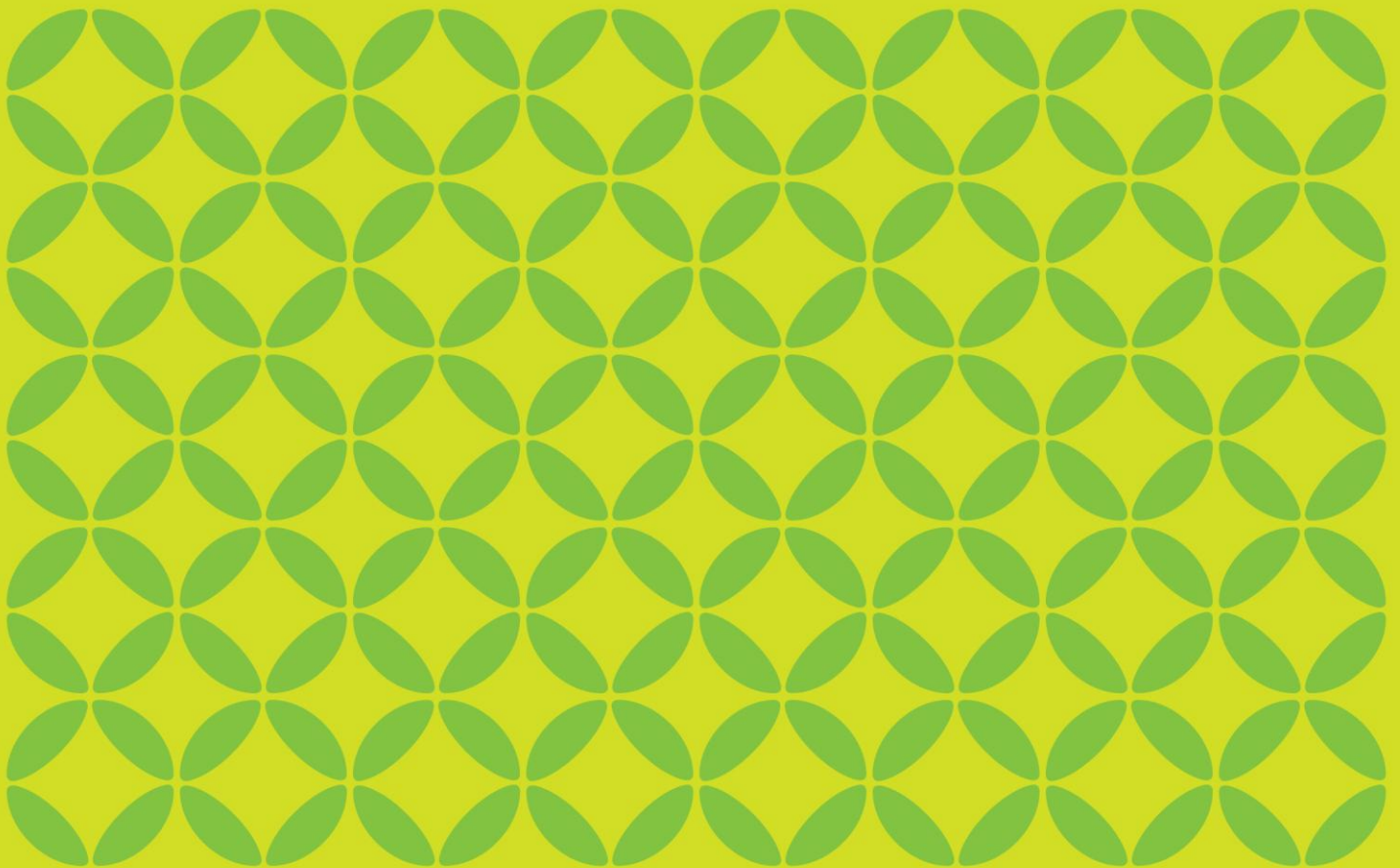


Upfront Carbon Reduction calculation guide

Guidance on calculation methods for the Upfront
Carbon Reduction and Impacts Disclosure credit in
Green Star Buildings v1.1



Version 1.1
25 May 2026

Version control

This guide applies to projects registered under version 1.1 of Green Star Buildings (all revisions). This guide is not applicable to projects targeting Green Star Buildings v1. These projects should refer to the *Upfront Carbon Emissions calculation guide v1*.

Release	Date	Description
1	25/05/2026	Initial release for Green Star Buildings v1.1.

As guidance in this guide is aligned with the NABERS Embodied Carbon rating tool, this guide may be updated if changes are made to NABERS. This guide will also be updated alongside the release of *Pathway C: NABERS Embodied Carbon pathway*.

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Acknowledgements

This guide was adapted from the *Upfront Carbon Emissions calculation guide – interim* developed in conjunction with thinkstep-anz. We thank them for their expertise, advice, and insights.

We want to thank our funding partners for the initial release, Lendlease, NSW Government and Sustainability Victoria.

Thanks also to the many individuals that provided input on the reference specifications.

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Introduction

About this guide

This document sets out the basis on which a project's upfront carbon emissions modelling should be completed for Green Star Buildings v1.1.

To help provide a higher level of comparability and certainty in upfront carbon calculation, this guide:

- Outlines the scope of inclusions and exclusions in calculating a project's upfront carbon emissions.
- Provides guidance on demonstrating reductions in upfront carbon emissions.
- Identifies the differences between comparing with a benchmark and a *Reference building*.
- Defines default *Reference building* materials.

This guide applies to version 1.1 of Green Star Buildings (all revisions). This guide is not applicable to projects targeting Green Star Buildings v1. These projects should refer to the *Upfront Carbon Emissions calculation guide v1*.

As guidance in this guide is aligned with the NABERS Embodied Carbon rating tool, this guide may be updated if changes are made to NABERS. This guide will also be updated alongside the release of *Pathway C: NABERS Embodied Carbon pathway*.

How to use this document

Upfront Carbon Reduction credit

The methodology in this guide is applied when targeting the *Upfront Carbon Reduction* credit in Green Star Buildings v1.1.

This guide applies to the two pathways available for the credit at the time of publication, *Pathway A: Benchmark pathway* and *Pathway B: Reference building pathway*.

Another pathway, *Pathway C: NABERS Embodied Carbon pathway*, is noted to be in development. This pathway will use a project's NABERS Embodied Carbon rating and will be available in future. Pathway A and Pathway B both reference the NABERS Embodied Carbon Rules for consistency across the pathways. More information can be found in Green Star Buildings v1.1 Submission Guidelines.

Guidance specific to the *Upfront Carbon Reduction* credit in Green Star, such as when the approach in Green Star differs to NABERS, is highlighted in green.

Impacts Disclosure credit

This credit encourages project teams to disclose the environmental impacts of the rated project across all life cycle modules determined through a life cycle assessment (LCA). Carbon emissions are one of the disclosed impacts alongside other environmental impacts such as photochemical ozone creation potential and eutrophication. This credit focuses on the declaration of the impacts to the GBCA and does not require a comparison with a *Reference building*.

The methodology in this guide is also applicable to projects targeting the *Impacts Disclosure* credit. Projects should refer to the *Calculating the Proposed building* section of the guide for the rules and scope that the life cycle assessment should follow in addition to the Submission Guidelines. References to calculating upfront carbon emissions for the *Proposed building* can be used to calculate the Climate Change impact category for modules A1 to A5 in the Life Cycle Assessment to be disclosed as part of the *Impacts Disclosure* credit.

Guidance specific to the *Impacts Disclosure* credit in Green Star is highlighted in blue.

Interdependence with other Green Star credits

Upfront Carbon Compensation

The building's upfront carbon emissions to be compensated for in this credit aligns with the upfront carbon emissions calculated in the *Upfront Carbon Reduction* credit.

Energy Use

The *Reference building* systems defined for the purposes of this credit should match the building fabric requirements for the *Reference building* systems as specified in the *Energy Use* credit.

Responsible Structure, Responsible Envelope, Responsible Systems and Responsible Finishes

Some information collated to calculate upfront carbon emissions may also assist project teams in making claims under the responsible products credits. In particular, Environmental Product Declarations, Climate Active certifications or similar are recognised under the Responsible Products Framework.

Applicability to Green Star project

Projects registered under Green Star Buildings v1.1 must use the version of the guide current at the time of registration or may choose to use a newer version at their discretion if available. Projects must use the selected version in its entirety.

Reference specifications should not change substantially between versions of this guide, however if you believe your project has been substantially affected by changes, please submit a Technical Question.

Feedback

GBCA welcomes feedback on the scope of the reference specifications and guidance provided. Feedback can be lodged as a feedback request in the [Green Star Manager](#).

Definitions

Benchmark

A defined carbon emissions intensity based on building type. This is used in *Pathway A: Benchmark pathway* as the baseline from which the *Proposed building* is compared against.

Embodied Carbon

Carbon emissions associated with materials and construction processes throughout the whole life cycle of a building. This includes Upfront Carbon, Use Stage Embodied Carbon, and End of Life Carbon, but not Operational Carbon.

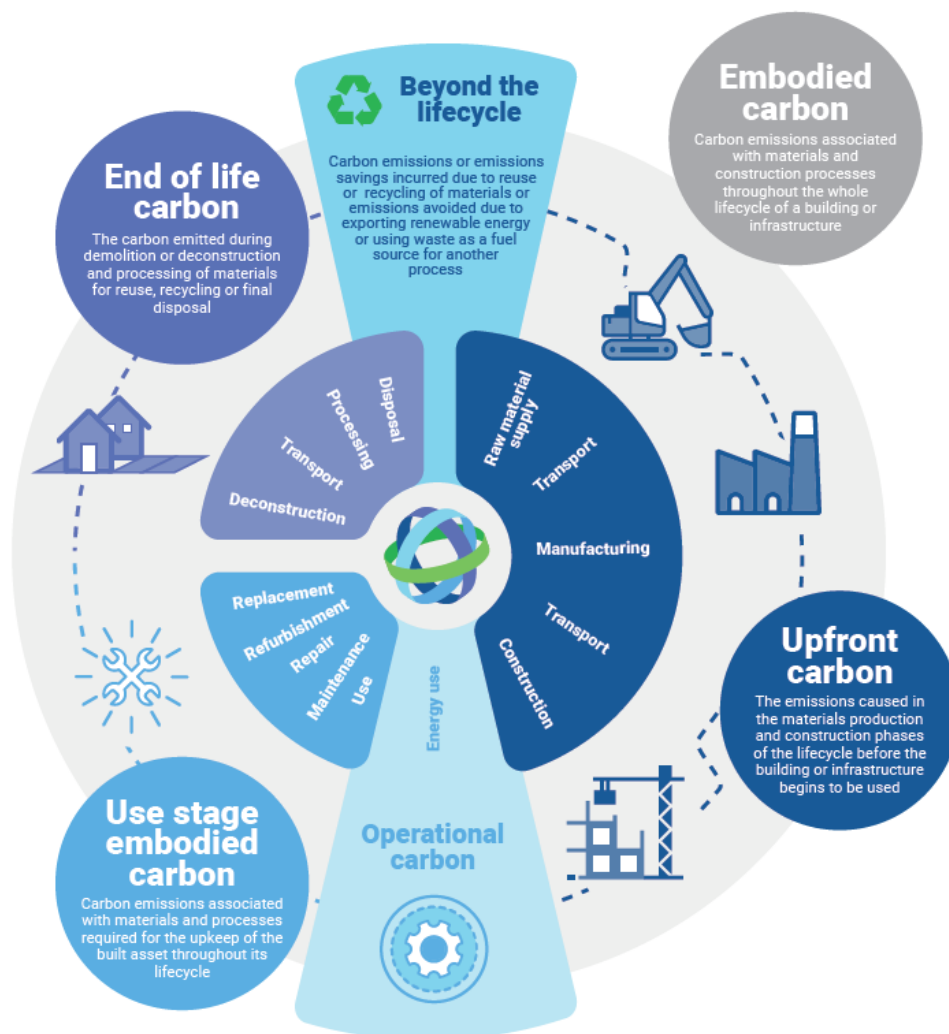


Figure 1 - Embodied carbon through a building's life cycle (World Green Building Council, 2019)

End of Life Carbon

The carbon emissions associated with deconstruction/demolition, transport from site, waste processing and disposal phases of a building's life cycle which occur after its use (Modules C1 to C4).

Life Cycle modules

The stages of a building's lifetime impacts as defined in EN15978 in Figure 2.

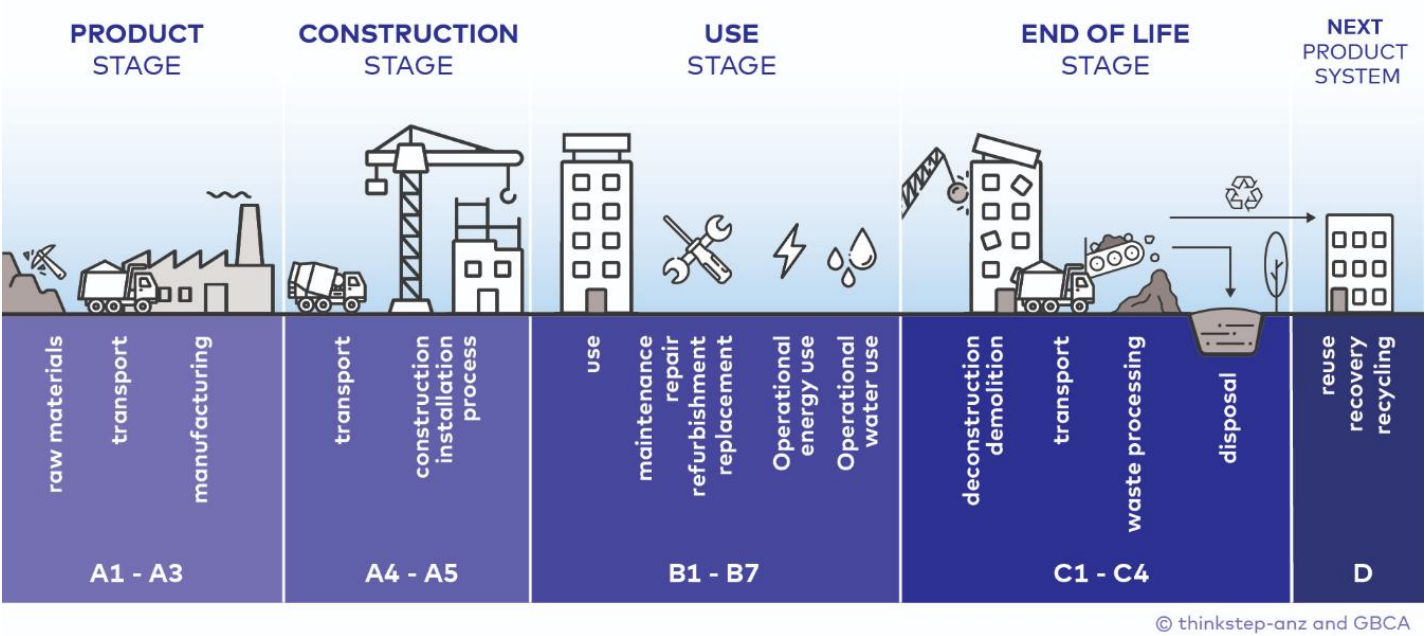


Figure 2 - Life cycle modules according to EN 15978

Operational Carbon

The emissions associated with energy used to operate the building (Module B6). Operational water (Module B7) is also often included as part of Operational Carbon.

Proposed building

The building works to be rated under the Green Star Buildings rating tool, as designed and modelled by the project team. This is relevant for both *Pathway A: Benchmark pathway* and *Pathway B: Reference building pathway*.

Reference building

A hypothetical project of the same size, shape, location, floor area and glazing areas as the *Proposed building* but is reflective of typical construction practices. This is used in *Pathway B: Reference building pathway* as the baseline from the *Proposed building* is compared against.

Upfront Carbon Emissions

The carbon emissions caused before the building begins to be used, i.e., during manufacture of building products, transport of building products to site and construction of the building (*EN 15978* Modules A1 to A5).

Use Stage Embodied Carbon

Emissions associated with materials and processes needed to maintain the building during use such as for maintenance, repair or refurbishments (*EN 15978* Modules B1 to B5).

Calculating upfront carbon emissions

There are two pathways available under the *Upfront Carbon Reduction* credit:

- *Pathway A: Benchmark pathway*
- *Pathway B: Reference building pathway*

Both pathways require the calculation of the upfront carbon emissions of the *Proposed building* however the baseline in which the *Proposed building* demonstrates a reduction on differs:

- *Pathway A: Benchmark pathway* – the baseline is a pre-defined benchmark defined by the GBCA.
- *Pathway B: Reference building pathway* – the baseline is a self-determined *Reference building* developed by the project team.

Calculating the *Proposed building*

The upfront carbon of the *Proposed building* must be calculated for both *Pathway A: Benchmark pathway* and *Pathway B: Reference building pathway*. Regardless of the pathway that is targeted, the scope and assumptions outlined in this section for the *Proposed building* remain the same.

Life cycle stages and system boundary

The system boundary follows *EN 15978:2011* and *EN 15804:2012+A2:2019*.

The following activities are **included**:

- Modules A1 to A3: Carbon emissions from the manufacture of products and materials used in the building.
- Module A4: Transport of building products to site.
- Module A5: Site preparation works and construction of the building. This includes:
 - On-site construction activities, such as operation of cranes, excavators, and building site services, and the manufacture, transport and disposal of any wasted building products.
 - Energy (primarily diesel and electricity) used in construction machinery or site offices on the building site.
 - Manufacture, transport, and end-of-life treatment of materials that become construction waste.
 - For greenfield developments, this module also includes the carbon impacts of land use change (measured against the state of the land at the date of purchase).

The following activities are **excluded**:

- Manufacture of machinery and other capital goods.
- **Transport of staff to and from the construction site.**
- Electricity used off-site for professional services.
- Pre-construction activities, such as land acquisition and design (Module A0).

Please note:

- Both lists are intended to be illustrative and are not exhaustive.
- The exclusions above are aligned with a process-based life cycle assessment approach, e.g., *PCR 2019:14 Construction Products* from the International EPD System (IEPDS 2022, section 4.3.2) as used within EPD Australasia.

Impacts Disclosure Guidance

In addition to the above, modules B to D in accordance with *EN 15804:2012+A2:2019* are included in the life cycle assessment for the *Impacts Disclosure* credit.

Calculation methodology

All carbon footprint calculations shall be performed using Global Warming Potential over a 100-year time horizon (GWP100) in line with *ISO 14067:2018*. The most recent characterisation factors from the Intergovernmental Panel on Climate Change (IPCC) should be used where possible. At the time of writing, the *IPCC Sixth Assessment Report (AR6)* contains the most recent factors (IPCC, 2021). However, GWP100 factors following older assessment reports may also be used.

Following *EN 15804:2012+A2:2019*, the total carbon footprint – **GWP-Total** – is the sum of three constituent parts:

- GWP-Fossil: Carbon footprint arising from fossil sources.
- GWP-Biogenic: Carbon footprint arising from biogenic sources (net of emissions and removals).
- GWP-LULUC: Carbon footprint due to land use and land use change.

Impacts Disclosure Guidance

In addition to the above, the following indicators defined by *EN 15804:2012+A2:2019* are to be disclosed as identified in the Submission Guidelines:

- Impact Categories
 - Stratospheric ozone depletion potential (kg CFC11 eq)
 - Acidification potential of land and water (mol H⁺ eq)
 - Eutrophication, marine (kg N eq)
 - Eutrophication, freshwater (kg P eq)
 - Eutrophication, terrestrial (mol N eq)
 - Photochemical ozone creation potential (kg NMVOC eq)
 - Abiotic depletion, minerals & metals (kg Sb eq)
 - Abiotic depletion, fossils (MJ, net calorific value)
- Resource Use
 - Net use of fresh water (m³)

Rated area

As per the NABERS Embodied Carbon Rules, the overall rated area is based on the Gross Floor Area (GFA).

The Australian Institute of Quantity Surveyors' definition of GFA is applied. GFA is the sum of "Fully Enclosed Covered Area" and "Unenclosed Covered Area", as defined below (AIQS, 2013, Appendix A, Part 2).

Fully Enclosed Covered Area (FECA)

The sum of all such areas at all building floor levels, including basements (except unexcavated portions), floored roof spaces and attics, garages, penthouses, enclosed porches and attached enclosed covered ways alongside buildings, equipment rooms, lift shafts, vertical ducts, staircases and any other fully enclosed spaces and usable areas of the building, computed by measuring from the normal inside face of exterior walls but ignoring any projections such as plinths, columns, piers and the like which project from the normal inside face of exterior walls.

It shall not include open courts, light wells, connecting or isolated covered ways and net open areas of upper portions of rooms, lobbies, halls, interstitial spaces and the like which extend through the storey being computed.

Unenclosed Covered Area (UCA)

The sum of all such areas at all building floor levels, including roofed balconies, open verandahs, porches and porticos, attached open covered ways alongside buildings, undercrofts and usable space under buildings, unenclosed access galleries (including ground floor) and any other trafficable covered areas of the building which are not totally enclosed by full height walls, computed by measuring the areas between the enclosing walls or balustrade (i.e. from the inside face of the UCA excluding the wall or balustrade thickness).

When the covering element (i.e. roof or upper floor) is supported by columns, is cantilevered or is suspended, or any combination of these, the measurements shall be taken to the edge of the paving or to the edge of the cover, whichever is the lesser.

UCA shall not include eaves overhangs, sun shading, awnings and the like where these do not relate to clearly defined trafficable covered areas, nor shall it include connecting or isolated covered ways.

Refer to Section 4 Rated area in the [NABERS Embodied Carbon Rules](#) for further details.

Building attributes

The building attributes included in the *Upfront Carbon Reduction calculator* must align with Section 5 *Building attributes* in the [NABERS Embodied Carbon Rules](#). This includes:

- Site location
- Building type
- Building dimensions
- Structural methods and materials.

Refer to Section 5 *Building attributes* in the [NABERS Embodied Carbon Rules](#) for further details.

Green Star Guidance

Mixed use projects, where more than 20% of the GFA is not the predominant building type, are to use the same pathway for the whole building.

For example, if the project has a mix of spaces that are applicable under Pathway A and not applicable under Pathway A then Pathway B must be used.

Land use and land use change

As per the NABERS Embodied Carbon Rules, emissions associated with land use and land use change are included if the project site was a greenfield site at the time of purchase.

Refer to Section 6 *Land use and land use change* in the [NABERS Embodied Carbon Rules](#) for further details.

Minimum material coverage

Scope

Refer to Section 7.3 *Material coverage for each building type* in the [NABERS Embodied Carbon Rules](#) for the materials that must be included for Class 5 office, Class 2 residential, Class 9a hospital, Class 9b public building, Class 9b residential aged care and retirement living, Class 6 retail and Class 7b industrial.

Green Star Guidance

For building types not included in the NABERS Embodied Carbon Rules, projects are to include the structure (substructure and superstructure), envelope and systems as a minimum. The excluded materials in Section 7.4 *Excluded materials* can be applied to these building types.

This differs from Green Star Buildings v1 where at least 90% of physical materials and 90% of financial value of building products are included.

For tenanted buildings, upfront carbon emission from fitout works are considered in the *Upfront Fitout Emissions* sector specific credit available on the Green Star resources portal.

Repurposed materials

Building materials that are reused on site are considered to have an embodied carbon of zero in the *Proposed building*. Emissions associated with additional activities such as reprocessing and transporting of materials are to be added.

Shared elements

Refer to Section 7.10 Shared elements between buildings in the [NABERS Embodied Carbon Rules](#) for the approach to be used to apportion upfront carbon for shared elements.

Green Star Guidance

All buildings within the precinct targeting a Green Star Buildings v1.1 rating must use the same allocation method for each shared element for consistency.

If the approach in the NABERS Embodied Carbon Rules cannot be applied, a Technical Question is recommended.

Material quantities

Green Star Guidance

Material quantities used in the upfront carbon emissions calculation for the *Proposed building* are based on **as built quantities** used in the building and its construction. This may include:

- As built bill of quantities
- Quantities summary from a material supplier
- Invoices
- Delivery dockets
- Purchase orders
- Tender stage bill of quantities with a schedule of contract variations
- As built drawings
- Building information modelling (BIM) data LOD 400 or greater (refer to page 43 in the NABERS Embodied Carbon Rules for further details)

Emission factors

Data hierarchy

The following preference hierarchy must be used for emission factors:

- Product-specific emission factor: Emission factors that apply to the specific product used in the building shall be used wherever they are available. These emission factors should be producer and region specific, and independently verified. This can be a product specific Environmental Product Declaration (EPD), Product Carbon Footprint (PCF) or a Climate Active Product Certification provided they meet the conditions outlined in *Section 9.4.3.1 Conforming EPDs*, *Section 9.4.4.1 Conforming carbon footprint declarations* and *Section 9.4.5 Climate Active Public Disclosure Statements* in the [NABERS Embodied Carbon Rules](#) respectively.
- Industry specific EPD: An industry specific EPD where the manufacturer has contributed data. The worst-case emission factor in the EPD is used. Refer to *Section 9.4.3.4 Industry average EPDs* in the [NABERS Embodied Carbon Rules](#) for more details.

Green Star Guidance

The subsequent emission factor sources in the hierarchy are for applicable to Green Star only.

- Generic value from a database: **The default emission factor** from the [NABERS National material emission factors database](#) or an appropriate generic material from an LCA tool may be selected. Process LCA data is preferred however hybrid LCA data may be used if unavailable. When targeting *Pathway B: Reference building pathway*, the same type of data must be used for the same element. Refer to the *Calculating reductions against a Reference building* section below for further details.
- Generic value from a global literature scan: Where neither of the previous two options are possible, a generic emission factor from literature may be used, using the worst value found for the correct product type.

GBCA notes that the emission factors preference hierarchy will be updated in the final version of this guide.

Impacts Disclosure Guidance

A different data source may be used for other impact categories compared to the Climate Change impact category if the data source used for the *Upfront Carbon Reduction* credit does not disclose this information. The above hierarchy should still be followed with preference for product-specific factors followed by an industry specific EPD, generic values from a database and then generic values from a global literature scan as a last option.

Carbon removals

Carbon removals including carbon offsets associated with products certified to be carbon neutral and stored biogenic carbon do not contribute to or reduce the project's upfront carbon.

Green Star Guidance

Carbon removals are reported separately in the *Upfront Carbon Compensation* credit. These are calculated in accordance with the [NABERS Embodied Carbon Rules](#). Refer to the Submission Guidelines for further details.

This differs from Green Star Buildings v1 where carbon neutral products and stored biogenic carbon products can be assumed to have no carbon.

Transport emissions

Transport emissions in the *Upfront Carbon Reduction calculator* follow the NABERS default transport model whereby transport distances are predefined based on material and site location and a default transport emission factor is applied. Refer to *Section 11.3 Default transport data* in the [NABERS Embodied Carbon Rules](#) for further details.

Where project-specific data is available, refer to *Section 11.4 Project-specific transport data* in the [NABERS Embodied Carbon Rules](#) for further details.

Construction emissions

Construction emissions include emissions associated with energy use during construction, waste generated on-site and the end-of-life management of that waste. The *Upfront Carbon Reduction calculator* includes a default rate for the noted emission sources.

Where project-specific data is available, refer to *Section 12.3.2 Project-specific construction and commissioning energy* and *Section 12.4 Waste* in the [NABERS Embodied Carbon Rules](#) for further details.

Green Star Guidance

Emissions from demolishing a previous building and clearing the site of rubble are part of the previous building's life cycle. To encourage the retention of as much of the existing building as possible, particularly prior to the building's end of life (usually considered to be 50 years as a minimum), the embodied carbon of the demolished elements of the existing building are to be compensated for in the *Existing Building Compensation* criterion.

Embodied carbon of the demolished portion

The embodied carbon of the demolished portion of the existing building is calculated by one of the following:

- Using the benchmarks in *Minimum Expectation* for the relevant building type and the GFA of the demolished portion.
- Using data or drawings of the existing building to determine the material quantities and the emission factors follow the data hierarchy in the *Calculating reductions against a Reference building* section.

Where there are several plausible emission factor values for a particular project type, the project team should take the most conservative (i.e., highest) value. As an example, the project team should assume that concrete mixes have 0% cement replacement.

Projects that require demolition of existing building elements as they have been identified to be not fit-for-purpose may exclude the embodied carbon of these building elements from the emissions to be compensated. Building elements are considered not fit-for-

purpose if they demonstrate a significant lack of Australian Building Code compliance such that the safety of occupants are compromised and are beyond repair even with strengthening (e.g., permanent damage has been caused due to fire). An existing building that does not suit the spatial needs of the *Proposed building* does not qualify as 'not fit-for-purpose'. This is to be disclosed and justified clearly in the submission and agreed upon via a Technical Question with GBCA.

Addressing emissions

Where the existing building is 30 years old or less, 100% of the embodied carbon emissions associated with the demolished portion are offset.

Where the existing building is between 30 to 50 years old, the embodied carbon emissions associated with the demolished portion that are offset are discounted at 10% for every two additional years past year 30. For example, demolishing a 36-year-old building would require offsets to cover $(100\% - (3 \times 10\%)) = 70\%$ of the calculated embodied emissions.

The purchased offsets are nature-based and produced by a GBCA recognised entity. Further information is available in *Climate Positive Buildings and our Net Zero Ambitions*.

Calculating reductions against a benchmark

In *Pathway A: Benchmark Pathway*, the upfront carbon emissions of the *Proposed building* are to be demonstrated to be less than the pre-defined benchmarks defined in the Submission Guidelines.

Benchmarks

Green Star Guidance

The benchmarks used in this pathway are based on a review of the data collected from Green Star Buildings v1 submissions to date. The benchmarks have been provided for the building classes where sufficient data was available and based on the average emissions intensity of the *Reference building* from a variety of submissions. An uplift factor was included to ensure projects continue to be ambitious in demonstrating upfront carbon reductions that align with the Intergovernmental Panel on Climate Change (IPCC) recommendations on the built environment.

Overtime, benchmarks for other building classes may become available based on the data that can be collected or will be superseded by *Pathway C: NABERS Embodied Carbon pathway*.

Project teams that wish to propose a benchmark for a building class type that does not have one prescribed may do so via a Technical Question. The Technical Question should include detailed as built data from at least three buildings that have similar building attributes as the *Proposed building* and are of the same building type. The as built data should disclose the material quantities and emission factors assumed and project teams should include a justification on how the nominated buildings are suitable for setting a benchmark for the project.

Calculating reductions against a *Reference building*

In *Pathway B: Reference building pathway*, the upfront carbon emissions for the *Proposed building* are to be demonstrated to be less than a *Reference building*. The project team is to develop a *Reference building* and complete an upfront carbon calculation of the *Reference building* for comparison.

Developing the *Reference building*

Green Star Guidance

A *Reference building* is a hypothetical project of the same size, shape, location, floor area and glazing areas as the *Proposed building*. The aim of the *Reference building* is to provide a point of comparison for reductions against, by developing a building that could have been built in place of the *Proposed building* provided no work was done to reduce its upfront carbon emissions.

Therefore, both the *Proposed building* and *Reference building* must share the same:

- [Life cycle stages and system boundary](#) (guidance as per the *Proposed building*)
- [Calculation methodology](#) (guidance as per the *Proposed building*)
- Rated area (additional guidance below)
- Building attributes (additional guidance below)
- [Land use and land use change](#) (guidance as per the *Proposed building*)
- Minimum material coverage (additional guidance below)

The *Reference building* can differ on the following from the *Proposed building* to show reductions in upfront carbon as shown in the following sections:

- Material quantities (additional guidance below)
- Emission factors (additional guidance below)

Rated area

Green Star Guidance

Both the *Reference* and *Proposed buildings* must have the same rated area. This means that the following attributes related to the scale of the building have the same:

- Gross Floor Area
- Floor plate area

Building attributes

Green Star Guidance

Both the *Reference* and *Proposed buildings* must have the same building attributes unless it is demonstrated that adjustments have been for the purposes of reducing upfront carbon. There is additional guidance in the [Improvements through iterative design processes](#) section below on how the *Reference building* can be adjusted to show improvements through design choices and material selection.

The building attributes that are to be same include:

- Site location including:
 - Underlying geology
 - Planning constraints
- Building type / function

- Building dimensions including:
 - Overall building height
 - Number of floors
 - Span
 - Tenant requirements, particularly the same Net Lettable Area
 - Orientation
- Structural requirements, such as:
 - Design life
 - Vertical and lateral design loading
 - Durability
 - Fire resistance

Attributes covered under *Section 5.5 Structural method and materials* in the NABERS Embodied Carbon Rules are not required to be the same in the *Reference* and *Proposed buildings*.

Minimum material coverage

The scope of the building to be included in the *Reference building* must be the same as the *Proposed building* therefore the guidance in the *Minimum material coverage* section for the *Proposed building* above is also applicable to the *Reference building*.

Repurposed materials

Green Star Guidance

Repurposed materials can be considered to have an embodied carbon of zero in the *Proposed building*. Project teams that reuse an existing building – in whole or in part – may assume new construction in the *Reference building* for the reused parts of the building(s).

New construction for reused parts of a *Reference building*, assumes a structure of the same size and function, with similar strength and volume, constructed with modern specifications. There is no requirement to assume code compliance.

Unique building elements

Additional guidance relevant for the *Proposed* and *Reference building* is noted in relation to solar PV, glazing and shading systems.

Solar PV

Solar PV is excluded from both the *Proposed* and *Reference building*.

Green Star Guidance

Glazing

The *Reference building* should assume double glazing for improved thermal performance. Where the *Proposed building* design specifies single glazing, the *Reference building* shall be assessed using single glazing, and consequently a reduction in upfront carbon cannot be claimed.

Shading Systems

Shading systems, such as external louvres or fins, are considered an optional design element for the purposes of the *Upfront Carbon Reduction* credit. As such, these systems may be included in the *Reference building* if a shading system has been included to achieve the modelled energy performance for the *Proposed building*. The shading arrangement is similar in both the *Reference* and *Proposed building*. Where shading systems have not been included in the *Proposed building's* design, it must not be included in the assessment of either.

Material quantities

Green Star Guidance

Building material quantities used for the *Reference building* must be calculated using the quantities calculated as per the Detailed Design / Technical Design stage, or where justified, an earlier stage (see guidance in the *Improvements through iterative design processes* section below).

The material choices for the *Reference building* shall be based on present-day, typical construction methods, not the worst available. These standard material choices are defined in *Defining materials for use in the Reference building* section. They are defined for different classes of buildings.

The project team must disclose the default material specifications that are used by completing the *Upfront Carbon modelling report template*. The project may vary the material choices in the default material specifications in the *Appendix* however the deviations are disclosed, an explanation is provided and it is peer reviewed by a relevant, independent experienced professional (refer to the Submission Guidelines for a definition). Where the deviations are driven by project requirements to satisfy additional criteria, such as government issued guidelines, this can be used to justify a deviation from the default material specifications without a peer review.

Improvements through iterative design processes

Green Star Guidance

The *Reference building* is a hypothetical building that represents standard contemporary construction and operation practices. As noted earlier in the guide, the *Reference building* must have the same size, scale, volume, and function.

However, to allow for improvements and reductions in upfront carbon through design efficiencies and better material selection, the *Reference building* is to be measured at the stage referred to as Detailed Design / Technical Design¹. Any changes from that point onwards are only considered for the *Proposed building*.

In some cases, the *Reference building* can be based on earlier stages in the design process. To be considered, clear evidence should exist to demonstrate that embodied carbon reduction was part of the decision-making process, and part of the reason for the changes. Examples of evidence include:

- An original contract specification that has been altered.
- A minuted / traceable decision-making process where the original design was altered to reduce the embodied carbon.

Emission factors

Data sources

Green Star Guidance

Project teams may choose to use one of the follow data sources as a priority to determine the emission factors of the materials in the *Reference building*.

- Product-specific emission factor (as per the [Proposed building](#))
- Industry specific EPD (as per the [Proposed building](#))
- The average emission factor from the [NABERS National material emission factors database](#).

There is no hierarchy for the three sources identified above however if a product-specific or industry specific emission factor is selected that is greater than the average emission factor of the same material in the NABERS National material emissions factors database, this must be justified in the *Upfront Carbon modelling report*.

¹ As defined in the Australian National Building Specification (NATSPEC, 2022) or Royal Institute of British Architects' Plan of Work 2020 (RIBA, 2020) respectively.

Where none of the above are available or suitable, an appropriate generic value from a database, such as from an LCA tool, or a generic value from a global literature scan (as per the [Proposed building](#)) may be used.

The data used for each building element must be from a similar source for both the *Proposed building* and the *Reference building*. For example, where a product-specific EPD is used for the structural steel in *Proposed building*, process LCA data, such as a product-specific EPD or industry specific EPD, is to be used for the *Reference building*. It is possible to use a mix of process LCA data and hybrid LCA data within the same study, but it must be used for different building elements (e.g., process LCA for structural framing and hybrid LCA for building services).

GBCA notes that the emission factors will be updated in the final version of this guide.

Defining materials for use in the *Reference building*

The tables in this section aim to show where current typical construction practices lie. They are to be used to calculate the *Reference building's* upfront carbon emissions. The default material type specified aims to represent the predominant material type used for a given building element on the Australian market at the time of publication. This is a deliberate simplification as many buildings will use a mixture of materials for the same building element.

The default specifications outlined in this section are included in the *Upfront Carbon modelling report* template. The project team may vary the material choices in the tables as buildings and all building sites can be different however the alternative must be justified and peer reviewed by a relevant, independent experienced professional. Refer to the *Material quantities* section of the guide for the *Reference building* and the Submission Guidelines for further information.

The structure of the tables is based on the building element categories in the *Elemental Standard Form of Cost Analysis, 4th Edition* from the Royal Institution of Chartered Surveyors (RICS, 2012).

For the purposes of this guide, the tables have been broken into four main categories based on four typical building typologies:

- **Mid- to high-rise buildings.** These buildings are typically **five or more storeys**. They are usually taller than they are wide. The primary structural system may be any combination of reinforced concrete, structural steel framing and mass timber. Building types in this category include office towers, residential apartment towers and large hospitals, for example National Construction Code (NCC) Class 2, 5 and 9 buildings. See *Table 1: Default specifications for mid- to high-rise buildings* for reference materials.
- **Warehouse-type buildings.** These buildings are typically single storey with large spans and a large interior volume. They are much wider than they are tall. They may be designed for internal vehicle operation, e.g., forklifts, lift trucks and/or trucks. They are typically constructed using a portal frame over a reinforced concrete slab. Building types in this category include warehouses, logistics depots, large industrial buildings, large supermarkets and other large open-plan retail sites, for example NCC Class 6, 7 and 8 buildings. See *Table 2: Default specifications for warehouse-type buildings* for reference materials.
- **Low-rise buildings.** These buildings are typically **one to four storeys**. They are usually smaller than warehouse-type buildings (above), but larger than residential-type buildings (below). They may have a combination of large open-plan areas and smaller enclosed areas. They may be constructed using a standard reinforced concrete frame or reinforced concrete tilt-up panels continuously supported on reinforced concrete strip footings. Building types in this category include shopping centres, indoor sports venues, schools, libraries, smaller supermarkets, smaller industrial buildings and smaller hospitals, for example NCC Class 6, 7 and 9 buildings. See *Table 3: Default specifications for low-rise buildings* for reference materials.
- **Residential-type buildings.** These buildings are typically **one to two storeys and have the smallest floor area of the four building types**. Their construction resembles a **detached residential house**. They are typically constructed of timber stud or cold-form steel framing on a reinforced concrete slab. Building types in this category include medical practices, school buildings and extensions to the building types above, for example NCC Class 2, 3, 6 and 9 buildings. See *Table 4: Default specifications for residential-type buildings* for reference materials.

Where a specific project is a combination of multiple building types (e.g., a retail complex which features both warehouse-type buildings and low-rise buildings), the most appropriate building type shall be used for each part of the project.

In the tables below, the following is used:

- **Virgin steel (primary steel)** refers to steel produced primarily from iron ore. The most common manufacturing route is to use a Blast Furnace to convert iron ore to pig iron and then a Basic Oxygen Furnace (BOF) to convert pig iron into steel. All virgin steel contains some recycled content, but virgin iron/steel makes up the bulk of the product.
- **Recycled steel (secondary steel)** refers to steel produced primarily from steel scrap. The most common manufacturing route is an Electric Arc Furnace (EAF). While steel scrap is the main raw material, other alloying elements – including virgin iron – may be used to achieve the desired alloy composition. As such, recycled steel does not always contain 100% recycled content. The emission factor used for the EAF's electricity should reflect the real electricity mix supplied to the furnace. If the source of the steel is unknown, assume production in Australia using average Australian grid electricity without Renewable Energy Certificates.
- **Virgin aluminium (primary aluminium)** refers to aluminium produced primarily from aluminium ore (bauxite). Production involves conversion of bauxite into alumina and then electrolysis of alumina to produce aluminium. The emission factor used for the smelter's electricity should reflect the real electricity mix supplied to the smelter. If the source of the aluminium is unknown, assume production in China using average Chinese grid electricity without Renewable Energy Certificates.

- **Recycled aluminium (secondary aluminium)** refers to aluminium produced from post-consumer recycled (secondary) sources. Aluminium scrap is put into a melting furnace and may then be further alloyed before being cast, extruded or rolled. Unlike steel, which always contains some recycled content, aluminium may have any recycled content from 0% to 100%.
- **Cement replacement** includes the use of any supplementary cementitious materials (SCMs) to replace ordinary Portland cement in concrete. These include, but are not limited to, fly ash, ground granulated blast-furnace slag, and silica fume. Additionally, there is ongoing interest in the use of alternative cement products (such as Limestone-Portland cement) to improve concrete performance while also lowering the carbon intensity of concrete mix design.

Projects that choose to use the average emission factors from the NABERS National material emission factors database should refer to the most similar material category of the same material type specified in the default reference materials. For example:

- **Concrete:** where cement replacement is specified, the emission factor for the same concrete strength is used (i.e., the emission factor for 40MPa concrete is used when 40MPa concrete with 20% cement replacement is specified).
- **Steel or aluminium:** where steel is specified as virgin or recycled steel, the emission factor for the same steel product is used (i.e., the emission factor for bar/reinforcing steel is used when bar/reinforcing made from conventional recycled steel is specified).
- **Mechanical, Electrical and Plumbing (MEP):** the emission factor for MEP can be used in lieu of defining the reference materials for mechanical, electrical and electrical building elements.

Table 1: Default specifications for mid- to high-rise buildings

Category 1	Category 2	Building element	Default reference materials
Substructure	Substructure	Foundation	Concrete: 50MPa with 20% cement replacement Reinforcing steel: Bar/mesh reinforcing made from conventional recycled steel at 70kg/m ³ for pad footings, 115kg/m ³ for pile caps and 230kg/m ³ for ground beams Design should align with recommendations provided by the project geotechnical engineer
		Piles	Concrete: 50MPa with 20% cement replacement or 0% cement replacement for aggressive soils Design as per the <i>Proposed building</i> . Soil aggressiveness is provided by the project geotechnical engineer
		Ground-bearing slab	Concrete: 32MPa with 20% cement replacement Reinforcing steel: Bar/mesh reinforcing made from conventional recycled steel at 50 kg/m ³ for slab on grade and 150 kg/m ³ for suspended ground slab
		Basement retaining walls	Concrete: 50MPa with 20% cement replacement Reinforcing steel: Bar/mesh reinforcing made from conventional recycled steel at 110 kg/m ³
Superstructure	Frame	Columns	Concrete: 50MPa with 30% cement replacement
		Beams	Reinforcing steel: Bar/mesh reinforcing made from conventional recycled steel at 250-350kg/m ³ for columns and 220kg/m ³ for beams

Category 1	Category 2	Building element	Default reference materials
			OR Structural steel: Universal beams/columns or welded beams/columns (hot rolled) from grade 300 virgin steel or hollow sections made from grade 350 to 400 virgin steel
	Suspended Floors	Banded slab	Banded slab generally for commercial office and healthcare Concrete: 40MPa with 20% cement replacement
		Flat slab	Reinforcing steel: Bar/mesh reinforcing made from conventional recycled steel at 90-110kg/m³
		Composite slab	OR Concrete: 40MPa with 0% cement replacement Post tensioning: Strand reinforcing made from virgin steel at 6kg/m² and bar/mesh reinforcing made from conventional recycled steel at 50kg/m³ Flat slab generally for residential Concrete: 40MPa with 20% cement replacement Reinforcing steel: Bar/mesh reinforcing made from conventional recycled steel at 80-100kg/m³ OR Concrete: 40MPa with 0% cement replacement Post tensioning: Strand reinforcing made from virgin steel at 5kg/m² and bar/mesh reinforcing steel at 35kg/m³ Composite slab Concrete: 40MPa with 20% cement replacement Structural decking: Structural decking made from virgin steel 1.0mm BMT (base metal thickness) Reinforcing steel: Bar/mesh reinforcing made from conventional recycled steel at 100-120kg/m³
	Roof	Banded slab	Banded slab generally for commercial office and healthcare Concrete: 40MPa with 20% cement replacement
		Flat slab	Reinforcing steel: Bar/mesh reinforcing made from conventional recycled steel at 90-110kg/m³
		Steel roof	OR Concrete: 40MPa with 0% cement replacement Post tensioning: Strand reinforcing made from virgin steel at 7kg/m² and bar/mesh reinforcing made from conventional recycled steel at 55kg/m³ Roof covering: 60mm concrete pavers or gravel ballast Flat slab generally for residential Concrete: 40MPa with 20% cement replacement

Category 1	Category 2	Building element	Default reference materials
			Reinforcing steel: Bar/mesh reinforcing made from conventional recycled steel at 80-100kg/m³ OR Concrete: 40MPa with 0% cement replacement Post tensioning: Strand reinforcing made from virgin steel at 6kg/m² and bar/mesh reinforcing made from conventional recycled steel at 40kg/m³ Roof covering: 60mm concrete pavers or gravel ballast Steel roof Framing: Cold rolled steel purlins made from grade 450 virgin steel Cladding: Pre-painted steel sheeting made from virgin steel 0.42mm or 0.48mm BMT
Stairs and ramps	Stairs and ramps		Concrete: 40MPa with 20% cement replacement Reinforcing steel: Bar/mesh reinforcing made from conventional recycled steel at 125kg/m³
External walls	Structural external walls		Concrete: 50MPa with 30% cement replacement Reinforcing steel: Bar/mesh reinforcing made from conventional recycled steel at 260kg/m³ for gravity walls and 300 – 350kg/m³ for shear walls Finish: Cement render OR Concrete blockwork: 190mm concrete blocks, core-filled with 20MPa grout and bar reinforcing made from conventional recycled steel at 30kg/m³ Finish: Cement render
	Non-structural external walls		Concrete blockwork: 190mm concrete blocks, core-filled with 20MPa grout and bar reinforcing made from conventional recycled steel at 15kg/m³ OR Framing: Cold rolled steel frame made from grade 450 virgin steel Cladding: Pre-painted aluminium sheeting made from virgin aluminium 3.0mm BMT or 110mm clay bricks OR Curtain wall (spandrel / non-vision sections): Frame made from powder coated, virgin aluminium with pre-painted aluminium sheeting made from virgin aluminium 3.0mm BMT and a galvanised steel backpan sheet made from virgin steel 0.9mm BMT.
Windows and external doors	Curtain wall / façade		Curtain wall (glass / vision sections): Frame made from powder coated, virgin aluminium with 100% double glazing (or equivalent NCC 2022 Section J compliant façade design)

Category 1	Category 2	Building element	Default reference materials
			External shading: Louvre made from powder coated, virgin aluminium mounted externally to a frame made from powder coated, virgin aluminium
	Internal walls and partitions	Structural internal walls	Concrete: 50MPa with 30% cement replacement Reinforcing steel: Bar/mesh reinforcing made from conventional recycled steel at 260kg/m ³ for gravity walls and 300 – 350kg/m ³ for shear walls
		Non-structural internal walls	All typologies Wall partitions: 13 mm plasterboard two sides over cold rolled steel frame made from grade 450 virgin steel OR Concrete blockwork: 190mm concrete blocks, core-filled with 20MPa grout and bar reinforcing made from conventional recycled steel at 15kg/m ³ Commercial office and healthcare Glazed partitions: Single-glazed window with aluminium frame OR Wall partitions: 13mm plasterboard two sides over cold rolled steel frame made from grade 450 virgin steel
Finishes	Wall finishes	Wall feature finishes	Bathroom tiles: 5mm or 10mm ceramic tiles
	Floor finishes	Floor finishes	Commercial office Carpet flooring for ~80% of area and vinyl flooring for ~20% of area Access flooring: Cement core, steel pedestal for ~80% of area Residential Carpet flooring for ~20-30% of area, ceramic tiles for ~20-40% of area and timber flooring for ~20-40% of area
	Ceiling finishes	Ceiling finishes	Commercial office Choose the most appropriate ceiling system: - Aluminium frame suspended with aluminium sheeting - Steel frame suspended with mineral tiles - Steel frame suspended with plasterboard Residential Steel frame suspended with plasterboard
Building services/MEP	Services	Plumbing	- Building plumbing - Hot water system (e.g., gas boiler) - Showers - Electric pump

Category 1	Category 2	Building element	Default reference materials
			• Water tank
		Mechanical	As per the GBCA Energy Use Calculation Guide for Green Star Buildings v1.1
		Fire	Fire services
		Electrical	• Cables (communication and network) • Lighting • BMS and energy monitoring • Site power and electrical connection
		Transportation	Lift(s)/elevator(s) Escalator(s) (if used)

Table 2: Default specifications for **warehouse-type buildings**

Category 1	Category 2	Building element	Default reference materials
Substructure	Substructure	Foundations	Concrete: 32MPa with 20% cement replacement Reinforcing steel: Fibre reinforcing made from conventional recycled steel at 25 kg/m ³ for pad footings. Bar/mesh reinforcing made from conventional recycled steel at 100 kg/m ³ for ground beams.
		Piles	Concrete: 40MPa with 20% cement replacement or 0% cement replacement for aggressive soils Design as per the <i>Proposed building</i> . Soil aggressiveness is provided by the project geotechnical engineer
		Ground-bearing slab and hardstand	Internal warehouse areas Concrete: Slab thickness of 175 mm at 40MPa with 10% cement replacement Reinforcing steel: Fibre reinforcing made from conventional recycled steel at 35 kg/m ³ . Office areas Concrete: Slab thickness of 120mm at 32MPa with 20% cement replacement Reinforcing steel: Bar/mesh reinforcing made from conventional recycled steel at 40 kg/m ³ Hardstand Concrete: Slab thickness of 180mm at 32MPa with 10% cement replacement Reinforcing steel: Bar/mesh reinforcing made from conventional recycled steel at 30 kg/m ³
		Underground rainwater tank (if fitted)	Concrete: 32MPa with 20% cement replacement Reinforcing steel: Bar/mesh reinforcing made from conventional recycled steel at 40 kg/m ³
Superstructure	Frame	Portal frame	Structural steel: Universal beams/columns or welded beams/columns (hot rolled) made from grade 300 virgin steel
	Suspended Floors	Composite slab	Concrete: 32MPa with 20% cement replacement Structural decking: Structural decking made from virgin steel 1.0mm BMT (base metal thickness) Reinforcing steel: Bar/mesh reinforcing made from conventional recycled steel at 50 kg/m ³

Category 1	Category 2	Building element	Default reference materials
	Roof	Steel Roof	Warehouse without temperature control Framing: Cold rolled steel purlins made from grade 450 virgin steel Cladding: Pre-painted steel sheeting made from virgin steel 0.42mm or 0.48mm BMT and translucent polycarbonate sheeting ~10% of the area Roof internal lining: Laminated aluminium foil sarking and safety steel wire mesh Warehouse with temperature control Structural insulated panels (SIP): Galvanised virgin steel cladding (inside and outside) 0.42 or 0.48mm BMT and expanded polystyrene (EPS) insulation. If fire-rated, use polyisocyanurate (PIR) or mineral wool insulation instead.
	External walls	Bulk cladding	Warehouse without temperature control Framing: Cold rolled steel girts made from grade 450 virgin steel Cladding: Pre-painted steel sheeting made from virgin steel 0.42mm BMT or pre-painted aluminium sheeting made from virgin aluminium 3.0mm BMT OR <u>Precast concrete panel</u> Concrete: 40MPa with 10% cement replacement Reinforcing steel: Bar/mesh reinforcing steel at 175 kg/m³ Lifting lugs and dowel connections to be included. Warehouse with temperature control Structural insulated panels (SIP): Galvanised virgin steel cladding (inside and outside) 0.42 or 0.48mm BMT and expanded polystyrene (EPS) insulation greater than 100mm thick. If fire-rated, use polyisocyanurate (PIR) or mineral wool insulation instead.
		Cladding for office areas and front façade	Framing: Cold rolled steel girts made from grade 450 virgin steel Cladding: Pre-painted steel sheeting made from virgin steel 0.42mm BMT or pre-painted aluminium sheeting made from virgin aluminium 3.0mm BMT or 9mm fibre cement board OR <u>Precast concrete panel</u> Concrete: 40MPa with 10% cement replacement Reinforcing steel: Bar/mesh reinforcing steel at 175 kg/m³ Lifting lugs and dowel connections to be included.
	Windows and external doors	Windows and external doors	Frame made from powder coated, virgin aluminium with double glazing Steel roller door(s)

Category 1	Category 2	Building element	Default reference materials
	Internal walls and partitions	Non-structural internal walls	Wall partitions: 13mm plasterboard two sides over cold rolled steel frame made from grade 450 virgin steel
Finishes	Wall finishes	Wall feature finishes	Bathroom tiles: 5mm or 10mm ceramic tiles
	Floor finishes	Floor finishes	Warehouse area N/A (polished concrete) Office area Carpet flooring for ~80% of area and vinyl flooring for ~20% of area
	Ceiling finishes	Ceiling finishes	Warehouse area N/A (exposed ceiling) Office area Choose the most appropriate ceiling system: - Aluminium frame suspended with aluminium sheeting - Steel frame suspended with mineral tiles - Steel frame suspended with plasterboard 13mm plasterboard
Building services/MEP	Services	Plumbing	- Building plumbing - Hot water system (e.g., gas boiler) - Showers - Electric pump (if needed) - Water tank (if needed)
		Mechanical	As per the GBCA Energy Use Calculation Guide for Green Star Buildings v1.1
		Fire	Fire services
		Electrical	- Cables (communication and network) - Lighting - BMS and energy monitoring - Site power and electrical connection

Table 3: Default specifications for **low-rise buildings**

Category 1	Category 2	Building element	Default reference materials
Substructure	Substructure	Foundations	Concrete: 40MPa with 20% cement replacement Reinforcing steel: Bar/mesh reinforcing made from conventional recycled steel at 70kg/m ³ for pad footings, 115kg/m ³ for pile caps and 230kg/m ³ for ground beams Design should align with recommendations provided by the project geotechnical engineer
		Piles	Concrete: 40MPa with 20% cement replacement or 0% cement replacement for aggressive soils Design as per the <i>Proposed building</i> . Soil aggressiveness is provided by the project geotechnical engineer
		Ground-bearing slab	Concrete: 32MPa with 20% cement replacement Reinforcing steel: Bar/mesh reinforcing made from conventional recycled steel at 50kg/m ³ for slab on grade and 150kg/m ³ for suspended ground slab
		Basement retaining walls	Concrete: 40MPa with 20% cement replacement Reinforcing steel: Bar/mesh reinforcing made from conventional recycled steel at 110kg/m ³
Superstructure	Frame	Columns	Concrete: 40MPa with 20% cement replacement
		Beams	Reinforcing steel: Bar/mesh reinforcing made from conventional recycled steel at 200 - 250kg/m ³ for columns and 200kg/m ³ for beams
	Suspended Floors	Banded slab	Banded slab generally for commercial offices, retail or healthcare
		Flat slab	Concrete: 40MPa with 20% cement replacement
		Composite slab	Reinforcing steel: Bar/mesh reinforcing made from conventional recycled steel at 90-110kg/m ³ OR Concrete: 40MPa with 0% cement replacement Post tensioning: Strand reinforcing made from virgin steel at 6kg/m ² and bar/mesh reinforcing made from conventional recycled steel at 50kg/m ³
			Flat slab generally for residential Concrete: 40MPa with 20% cement replacement Reinforcing steel: Bar/mesh reinforcing made from conventional recycled steel at 80-100kg/m ³ OR Concrete: 40MPa with 0% cement replacement

Category 1	Category 2	Building element	Default reference materials
			Post tensioning: Strand reinforcing made from virgin steel at 5kg/m² and bar/mesh reinforcing made from conventional recycled steel at 35kg/m³ Composite slab Concrete: 40MPa with 20% cement replacement Structural decking: Structural decking made from virgin steel 1.0mm BMT (base metal thickness) Reinforcing steel: Bar/mesh reinforcing made from conventional recycled steel at 100-120kg/m³
Roof	Banded slab		Banded slab Concrete: 40MPa with 20% cement replacement Reinforcing steel: Bar/mesh reinforcing made from conventional recycled steel at 90-110kg/m³ OR Concrete: 40MPa with 0% cement replacement Post tensioning: Strand reinforcing made from virgin steel at 7kg/m² and bar/mesh reinforcing made from conventional recycled steel at 55kg/m³
	Flat slab		Flat slab Concrete: 40MPa with 20% cement replacement Reinforcing steel: Bar/mesh reinforcing made from conventional recycled steel at 80-100kg/m³ OR Concrete: 40MPa with 0% cement replacement Post tensioning: Strand reinforcing made from virgin steel at 6kg/m² and bar/mesh reinforcing made from conventional recycled steel at 40kg/m³
	Steel roof		Steel roof Framing: Cold rolled steel purlins made from grade 450 virgin steel Cladding: Pre-painted steel sheeting made from virgin steel 0.42mm or 0.48mm BMT
Stairs and Ramps	Stairs and Ramps		Concrete: 40MPa with 20% cement replacement Reinforcing steel: Bar/mesh reinforcing made from conventional recycled steel at 125 kg/m³
External structural walls	Structural external walls		Concrete: 50MPa with 30% cement replacement Reinforcing steel: Bar/mesh reinforcing made from conventional recycled steel at 180kg/m³ for gravity walls and 250 – 300kg/m³ for shear walls If precast, lifting lugs and dowel connectors to be included. Finish: Cement render

Category 1	Category 2	Building element	Default reference materials
			OR Concrete blockwork: 190mm concrete blocks, core-filled with 20MPa grout and bar reinforcing made from conventional recycled steel at 30kg/m³ Finish: Cement render
		Non-structural external walls	Concrete blockwork: 190mm concrete blocks, core-filled with 20MPa grout and bar reinforcing made from conventional recycled steel at 15kg/m³ OR Framing: Cold rolled steel frame made from grade 450 virgin steel Cladding: Pre-painted aluminium sheeting made from virgin aluminium 3.0mm BMT or 110mm clay bricks OR Curtain wall (spandrel / non-vision sections): Frame made from powder coated, virgin aluminium with pre-painted aluminium sheeting made from virgin aluminium 3.0mm BMT and a galvanised steel backpan sheet made from virgin steel 0.9mm BMT.
	Windows and external doors	Windows and external doors	Curtain wall (glass / vision sections): Frame made from powder coated, virgin aluminium with 100% double glazing (or equivalent NCC 2022 Section J compliant façade design) External shading: Louvre made from powder coated, virgin aluminium mounted externally to a frame made from powder coated, virgin aluminium Steel roller door(s)
	Internal walls and partitions	Non-structural internal walls	Wall partitions: 13mm plasterboard two sides over cold rolled steel frame made from grade 450 virgin steel
	Wall finishes	Wall feature finishes	Bathroom tiles: 5mm or 10mm ceramic tiles
Finishes	Floor finishes	Floor finishes	Commercial office Carpet flooring for ~80% of area and vinyl flooring for ~20% of area Residential Carpet flooring for ~20-30% of area, ceramic tiles for ~20-40% of area and timber flooring for ~20-40% of area
	Ceiling finishes	Ceiling finishes	Commercial office Choose the most appropriate ceiling system: - Aluminium frame suspended with aluminium sheeting - Steel frame suspended with mineral tiles - Steel frame suspended with plasterboard Residential

Category 1	Category 2	Building element	Default reference materials
Building services/MEP	Services		Steel frame suspended with plasterboard
		Plumbing	• Building plumbing • Hot water system (e.g., gas boiler) • Showers • Electric pump (if needed) • Water tank (if needed)
		Mechanical	As per the GBCA Energy Use Calculation Guide for Green Star Buildings v1.1
		Fire	Fire services
		Electrical	• Cables (communication and network) • Lighting • BMS and energy monitoring • Site power and electrical connection
		Transportation	• Lift(s)/elevator(s) • Escalator(s) (if used)

Table 4: Default specifications for residential-type buildings

Category 1	Category 2	Building element	Default reference materials
Substructure	Substructure	Foundation	Concrete: 32MPa with 20% cement replacement Reinforcing steel: Bar/mesh reinforcing made from conventional recycled steel at 70kg/m³ for pad footings and 230kg/m³ for ground beams Design should align with recommendations provided by the project geotechnical engineer
		Piles	Concrete: 40MPa with 20% cement replacement or 0% cement replacement for aggressive soils Design as per the <i>Proposed building</i>. Soil aggressiveness is provided by the project geotechnical engineer
		Ground-bearing slab	Concrete: 32MPa with 20% cement replacement Reinforcing steel: Bar/mesh reinforcing made from conventional recycled steel at 40kg/m³
Superstructure	Suspended Floors	Framing	Treated engineered softwood timber or laminated veneer lumber (LVL) beams and joists or cold rolled steel joists made from grade 450 virgin steel
	Roof	Roof	Framing: Treated softwood timber or cold rolled steel trusses Cladding: Pre-painted steel sheeting made from virgin steel 0.42mm BMT for 50% of area and concrete roof tiles for 50% of area
	Stairs and Ramps	Stairs and Ramps	Timber: Treated engineered softwood timber or laminated veneer lumber (LVL) stringers and treads OR Concrete: 40MPa with 20% cement replacement Reinforcing steel: Bar reinforcing steel made from conventional recycled steel at 125 kg/m³
	External walls	External walls	Framing: Treated softwood timber stud frame or cold rolled steel frame Cladding: 110mm clay bricks for ~33% of the area, pre-painted steel sheeting made from virgin steel 0.42mm BMT for ~33% of area and 9-15mm thick fibre cement board for ~33% of area OR Concrete: 50MPa with 30% cement replacement Reinforcing steel: Bar/mesh reinforcing made from conventional recycled steel at 180kg/m³ for gravity walls and 250 – 300kg/m³ for shear walls

Category 1	Category 2	Building element	Default reference materials
	Windows and external doors	Windows and external doors	Curtain wall (glass / vision sections): Frame made from powder coated, virgin aluminium with 100% double glazing External shading: Louvre made from powder coated, virgin aluminium mounted externally to a frame made from powder coated, virgin aluminium
	Internal walls and partitions	Non-structural internal walls	Wall partitions: 13mm plasterboard two sides over treated softwood timber stud frame or cold rolled steel frame made from grade 450 virgin steel
Finishes	Wall finishes	Wall feature finishes	Bathroom tiles: 5mm or 10mm ceramic tiles
	Floor finishes	Floor finishes	Carpet flooring for ~80% of area and vinyl flooring for ~20% of area
	Ceiling finishes	Ceiling finishes	Ceiling covering: 10mm plasterboard
Building services/MEP	Services	Plumbing	• Building plumbing • Hot water system (e.g., gas boiler) • Showers • Electric pump (if needed) • Water tank (if needed)
		Mechanical	As per the GBCA Energy Use Calculation Guide for Green Star Buildings v1.1
		Fire	Fire services
		Electrical	• Cables (communication and network) • Lighting • BMS and energy monitoring • Site power and electrical connection
		Transportation	• Lift(s)/elevator(s) • Escalator(s) (if used)

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